山东建筑大学　　计算机学院　　实验报告

班级：　软件 212　姓名：　孙帅康　　学号：202111110337　　实验成绩：

课程：　　同组者：尹皓琪，王潇文，邵亚琦，姜魁哲　实验日期：2024.3.28

实验四 负载均衡

一、实验目的

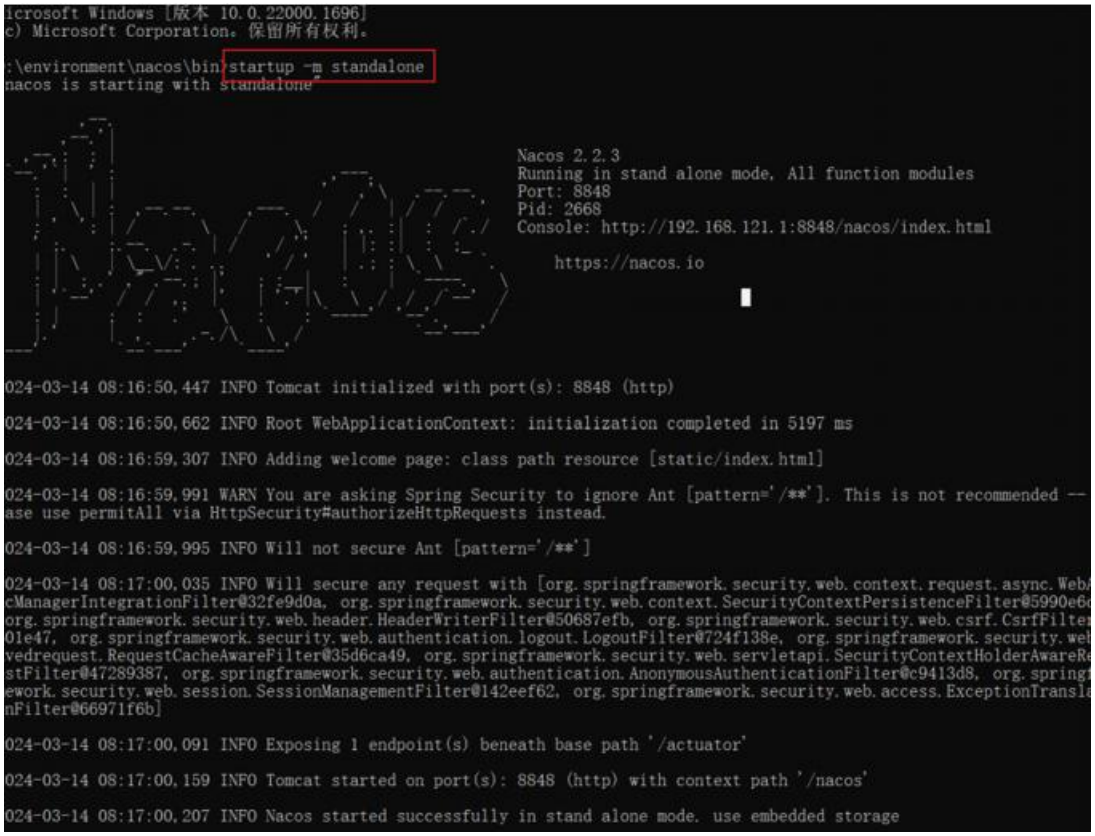
掌握负载均衡的配置实现方法。

二、实验内容

通过 OpenFeign 实现远程调用

通过 Openfeign 实现服务调用

1. 启动 nacos



2. 建立项目，对 pom.xml 文件进行配置

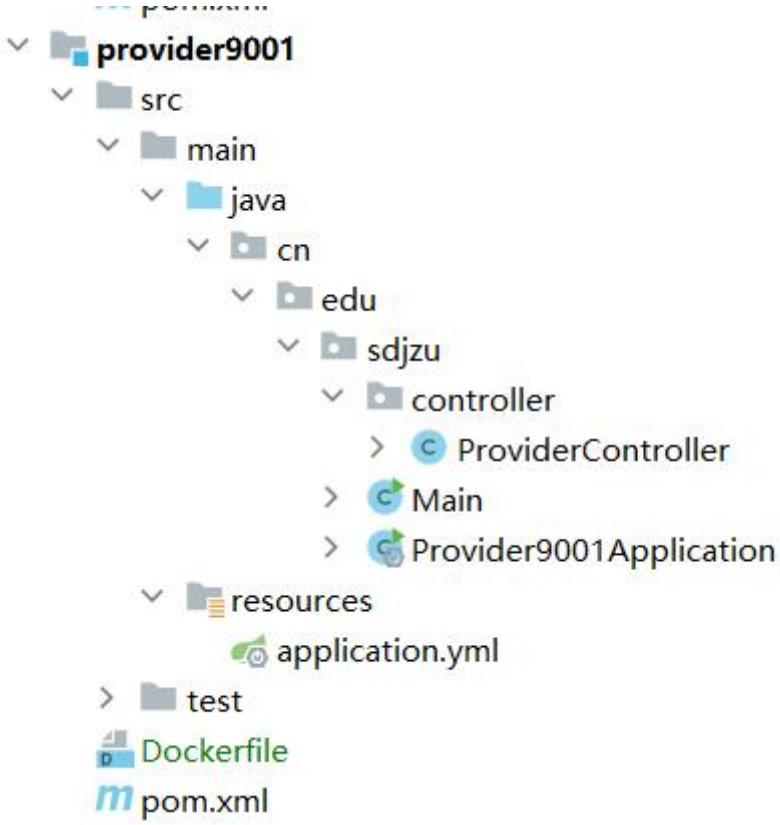
添加依赖

```
<dependencyManagement>
    <dependencies>
        <dependency>
            <groupId>org.springframework.boot</groupId>
            <artifactId>spring-boot-dependencies</artifactId>
            <version>3.0.2</version>
            <type>pom</type>
            <scope>import</scope>
```

```
            </dependency>

            <dependency>
                <groupId>org.springframework.cloud</groupId>
                <artifactId>spring-cloud-dependencies</artifactId>
                <version>2022.0.0</version>
                <type>pom</type>
                <scope>import</scope>
            </dependency>

            <dependency>
                <groupId>com.alibaba.cloud</groupId>

<artifactId>spring-cloud-alibaba-dependencies</artifactId>
                <version>2022.0.0.0</version>
                <type>pom</type>
                <scope>import</scope>
            </dependency>

            <dependency>
                <groupId>com.mysql</groupId>
                <artifactId>mysql-connector-j</artifactId>
                <version>8.3.0</version>
            </dependency>

            <dependency>
                <groupId>org.mybatis.spring.boot</groupId>
                <artifactId>mybatis-spring-boot-starter</artifactId>
                <version>3.0.3</version>
            </dependency>

            <dependency>
                <groupId>org.projectlombok</groupId>
                <artifactId>lombok</artifactId>
                <version>1.18.24</version>
            </dependency>
        </dependencies>
    </dependencyManagement>



    <build>
```

```
        <plugins>
            <plugin>
                <groupId>org.springframework.boot</groupId>
                <artifactId>spring-boot-maven-plugin</artifactId>
                <configuration>
                    <excludes>
                        <exclude>
                            <groupId>org.projectlombok</groupId>
                            <artifactId>lombok</artifactId>
                        </exclude>
                    </excludes>
                </configuration>
            </plugin>
        </plugins>
    </build>
```

3. 在项目下建立模块 module　命名为 provider9001



4. 对 provider9001 的 pom 文件进行配置

添加依赖：

```
<dependencies>
        <dependency>
```

```
            <groupId>com.alibaba.cloud</groupId>

<artifactId>spring-cloud-starter-alibaba-nacos-discovery</artifac
tId>

        </dependency>
        <dependency>
            <groupId>org.springframework.boot</groupId>
            <artifactId>spring-boot-starter-actuator</artifactId>

        </dependency>
        <dependency>
            <groupId>org.springframework.boot</groupId>
            <artifactId>spring-boot-starter-web</artifactId>
            <version>3.0.2</version>
        </dependency>
    </dependencies>
```

5.在 resources 文件夹下建立 application.yml 文件

```
server:
  port: 9001
spring:
  application:
    name: provider9001
  cloud:
    nacos:
      discovery:
        server-add: 192.168.23.1:8848
        service: provider
management:
  endpoints:
    web:
      exposure:
        include: '*'
```

6.建立主程序 Provider9001Application.java 文件

```
package cn.edu.sdjzu;

import
org.springframework.boot.autoconfigure.SpringBootApplication;
import org.springframework.boot.builder.SpringApplicationBuilder;
@SpringBootApplication
```

```
public class Provider9001Application {
    public static void main(String[] args){

        new
SpringApplicationBuilder(Provider9001Application.class).run(args);
    }

}
```

7.新建包controller在包下建立ProviderController.java文件

```
package cn.edu.sdjzu.controller;



import org.springframework.web.bind.annotation.GetMapping;
import org.springframework.web.bind.annotation.RestController;
import org.springframework.web.bind.annotation.RequestMapping;
import org.springframework.web.bind.annotation.RequestMethod;
@RestController
public class ProviderController {
     @RequestMapping(value="/hello",method = RequestMethod.GET)
    public String hello(){
        return "hello world,provider9001";
    }
}
```

8.用同样的方法步骤建立provider9002模块

建立新的module　命名为consumer9004

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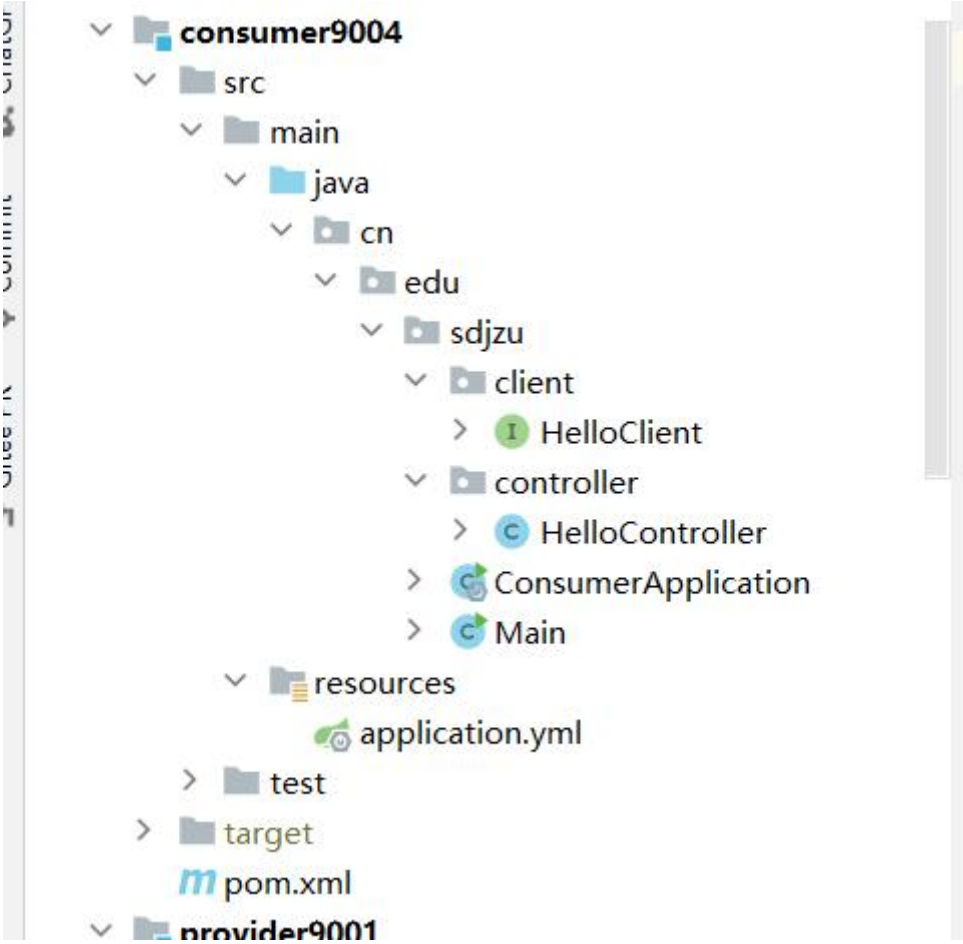


1. 对 pom 文件进行配置

```
<dependencies>
        <dependency>
            <groupId>com.alibaba.cloud</groupId>
            <artifactId>spring-cloud-starter-alibaba-nacos-discov
ery</artifactId>

        </dependency>
        <dependency>
            <groupId>org.springframework.boot</groupId>
            <artifactId>spring-boot-starter-actuator</artifactId>

        </dependency>
        <dependency>
            <groupId>org.springframework.boot</groupId>
            <artifactId>spring-boot-starter-web</artifactId>
            <version>3.0.2</version>
        </dependency>
```

```
        <dependency>
            <groupId>org.springframework.cloud</groupId>
            <artifactId>spring-cloud-starter-loadbalancer</artifa
ctId>
        </dependency>
        <dependency>
            <groupId>org.springframework.cloud</groupId>
            <artifactId>spring-cloud-starter-openfeign</artifactI
d>
        </dependency>
    </dependencies>
```

2.在 resources 文件夹下建立 application.yml 文件

```
server:
  port: 9004
spring:
  application:
    name: consumer9004
  cloud:
    nacos:
      discovery:
        server-add: 192.168.23.1:8848
        service: ${spring.application.name}
management:
  endpoints:
    web:
      exposure:
        include: '*'
```

3.建立主程序 ConsumerApplication.java 文件

```
package cn.edu.sdjzu;

import org.springframework.boot.SpringApplication;
import org.springframework.boot.autoconfigure.SpringBootApplicati
on;
import org.springframework.cloud.openfeign.EnableFeignClients;

@SpringBootApplication
@EnableFeignClients

public class ConsumerApplication {
    public static void  main(String[] args){
```

```
        SpringApplication.run(ConsumerApplication.class,args);
    }
}
```

4.建立包 client 在包下建立接口文件 HelloClient

```
package cn.edu.sdjzu.client;

import org.springframework.cloud.openfeign.FeignClient;
import org.springframework.web.bind.annotation.GetMapping;

@FeignClient("provider")
public interface HelloClient {
    @GetMapping("/hello")
   String sayHello();

}
```

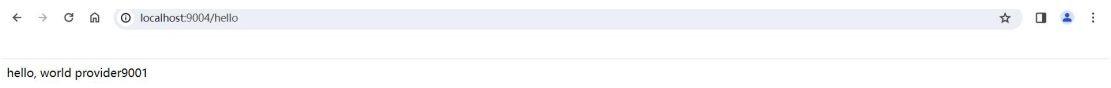
5.建立包 controller 在包下建立 HelloController.java 文件

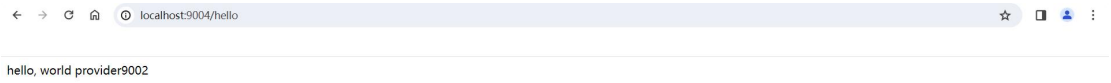
```
package cn.edu.sdjzu.controller;

import cn.edu.sdjzu.client.HelloClient;
import org.springframework.beans.factory.annotation.Autowired;
import org.springframework.web.bind.annotation.GetMapping;
import org.springframework.web.bind.annotation.RestController;

@RestController
public class HelloController {
    @Autowired
   private HelloClient helloClient;
    @GetMapping("/hello")
   public String sayHello(){
       return helloClient.sayHello();
   }

}
```

运行 provider9001，和 provider9002，consumer9004 文件

6. 结果显示：

点击刷新，会交错显示





三、实验总结

通过本次实验学会了通过 OpenFeign 实现负载均衡，注意调用的方法与注解。